VISHWAKARMA AWARDS
A Maker Bhavan Foundation Initiative

MAKER BHAVAN FOUNDATION

VISHWAKARMA AWARDS 2026-27

DERMASENSE

VK26-IITI-HT-0050

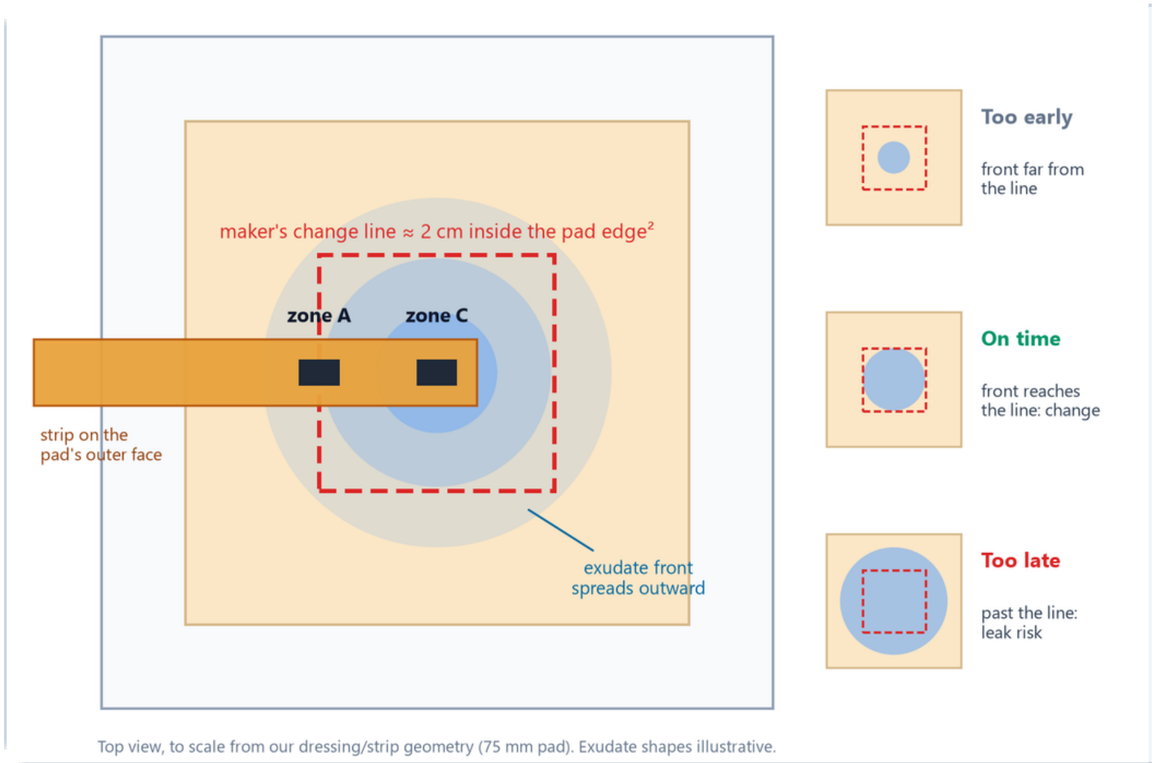
Health Tech · Emergency Response & Remote Monitoring

Team: Soham Prabhakar Kadu · Bhakti Ajay Kadam

PROBLEM CONTEXT

EVIDENCE FROM PUBLISHED SOURCES

44.9 %	≈ 2 cm · 75 %	1 point	> 10×
DRESSING CHANGES MADE WHILE MOISTURE WAS OPTIMAL[1]	SOME DRESSINGS SHOW WHEN IT IS TIME TO CHANGE [2, 14]	ONE EXISTING SENSOR TAKES A SINGLE-POINT READING[1]	REPORTED EXUDATE AMOUNTS VARY BY >10×[6]



Our illustration shows why the change line can be hidden under a cover (drawn to scale from our strip design).

The problem and who we designed for

- Problem: a carer cannot see whether a covered foam dressing has reached its maker's change line.
- Result: early changes mean needless pain and waste; late changes risk strike-through leakage [2] and periwound maceration (moisture-damaged skin) [21].
- Who: nurses and family carers supporting people with draining chronic wounds, including diabetic foot, leg and pressure ulcers.
- We will: show when that change line is reached, from outside, without lifting the dressing.

Project at a glance: Dermasense · Team VK26-IITI-HT-0050 · Health Tech: Emergency Response & Remote Monitoring · Our idea: read a dressing's change line from outside, without lifting it.

What carers can use today — and what is still missing

EARLY SEARCH · NOT EXHAUSTIVE

Current options and their limits

Today's option	Without lifting?	Maker's rule?	What limits it
Fixed schedule (clock)	✗	✗	hard to predict because fluid amounts vary >10×[6]
Lift and look	✗	✓	requires removing the dressing; removal can hurt[20]
Visual change indicator[14]	✗	✓	only works when the indicator is visible
Single-point moisture sensor[1]	◐	✗	measures moisture at one point
Closed-loop smart bandage[5]	✓	✗	tested in mice; active therapy adds cost and a regulatory burden [5]
Diaper wetness strip[4]	✓	✗	designed for diapers, not wound dressings
Dermasense (proposed)	✓	✓	**not yet tested; E2 and E4 will evaluate it**

✓ yes ✗ no ◐ partly. Our search of papers, patents and products was not exhaustive.

What we searched and what we are testing

- Searched: papers, patents and products [1, 3, 4, 5, 14].
- Found: in our search, no low-cost sensor reads a dressing's own change line.
- Our focus: show when fluid reaches that change line, without lifting the dressing.
- Why now: carbon can be laser-written on polyimide in one pass [7], and dressing makers already define a change line [2, 14].
- Not our goal: diagnosis or treatment. We are exploring a way to report how full the dressing is.

Question we are trying to answer: How can a carer tell when a covered dressing needs changing without lifting it?

User Journey: Before State

CURRENT CARE · PUBLISHED EVIDENCE

1.9–4.5 per 1,000	101 million	19–34 %	40.3 %
CHRONIC WOUND PREVALENCE IN TWO INDIAN SURVEYS [16, 17]	PEOPLE IN INDIA LIVING WITH DIABETES[18]	ESTIMATED LIFETIME FOOT-ULCER RISK WITH DIABETES[19]	PATIENTS SAY DRESSING-CHANGE PAIN IS THE WORST PART[20]

Illustrative persona, not a real patient: a person with a diabetic foot ulcer living at home with family support and periodic nursing visits.

STAGE	1 Dressing is applied	2 Dressing fills over time	3 Carer decides	4 Change	5 Repeat
WHAT HAPPENS	A foam dressing sits under a bandage or sock	Fluid builds up beneath the cover	Follow a schedule, take a look, or wait for leakage	Too early or too late	About five visits a week[13]
PAIN POINT	The change line is hidden under the cover	There is no clear signal	The decision is uncertain	Painful removal and a wasted dressing, or leakage and skin maceration	About 18 minutes per visit[13]

Illustrative care journey based on published data; the persona is not a real patient.

What published studies tell us

- Time: One UK patient survey reported about 18 minutes per dressing visit and five visits a week [13].
- Waste: In one study, 44.9% of recorded changes happened while moisture was still in the optimal range [1].
- Cost: Indian retail listings show ₹242–514 for one 10 × 10 cm foam dressing [15]; every premature change discards one.
- Safety: a dressing left on too long can leak and macerate the surrounding skin [21]; removal can hurt [20].
- Who carries it: the patient, family carer and nurse all manage this routine.

Insight: when a dressing is covered, a carer cannot see whether it has reached its change line.

User Journey: After State (Intended)

DESIGN INTENT · NOT CLINICALLY TESTED

0 lifts	up to 44.9 %	≤ ₹100	2–5 strips
DESIGN GOAL: CHECK WITHOUT LIFTING	EARLY CHANGES A RELIABLE ALERT COULD AVOID[1]	TARGET STRIP COST (ESTIMATE)	PAID BY ONE AVOIDED CHANGE (PROJECTION)[15]

The same illustrative persona with the proposed workflow. This has not been clinically tested.

STAGE	1 Strip on	2 Reader on	3 Check status	4 Change line reached	5 Nearby check
WHAT HAPPENS	placed on the pad's outer face	clips on: 4 pins, 2 dowels	Green OK, without lifting the dressing	Amber CHANGE; red CHECK for a possible leak	A nearby phone reads the status over Bluetooth (hardware test pending)
GAIN	**nothing on the wound**	**same routine**	**A clearer decision, if testing supports it**	**A prompt to check the dressing**	**Status checked remotely, without disturbing the wrap**

Intended workflow only. Bench testing comes first; the device has not been tested on people.

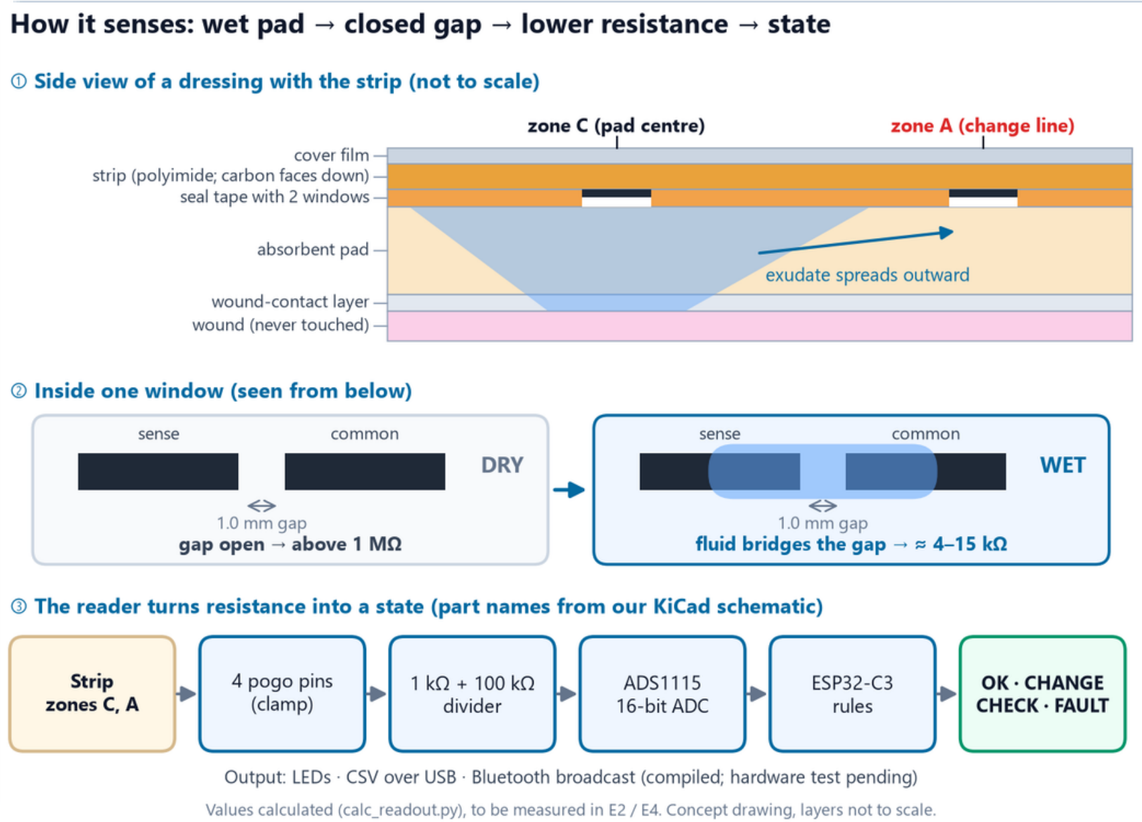
What each person could gain

- Patient: fewer needless removals, the step patients find most painful [20].
- Family carer: a clear OK / CHANGE signal instead of guesswork.
- Nurse: checks the status in seconds without unwrapping the bandage (design goal).
- Budget: reusable reader ≈ ₹560–1,160 (BOM estimate) + strip ≤ ₹100 (target); one avoided ₹242–514 dressing change pays for 2–5 strips (projection).

Design Principle: The intended setup keeps the usual care routine. The strip sits on the dressing's outer face; the reader is reused. E4 will test the ±10% alert target.

System Architecture & Specs

DESIGN TARGETS · NOT YET BENCH-TESTED



Concept drawing: how a moisture reading could become a status. Values are design targets, not measured results.

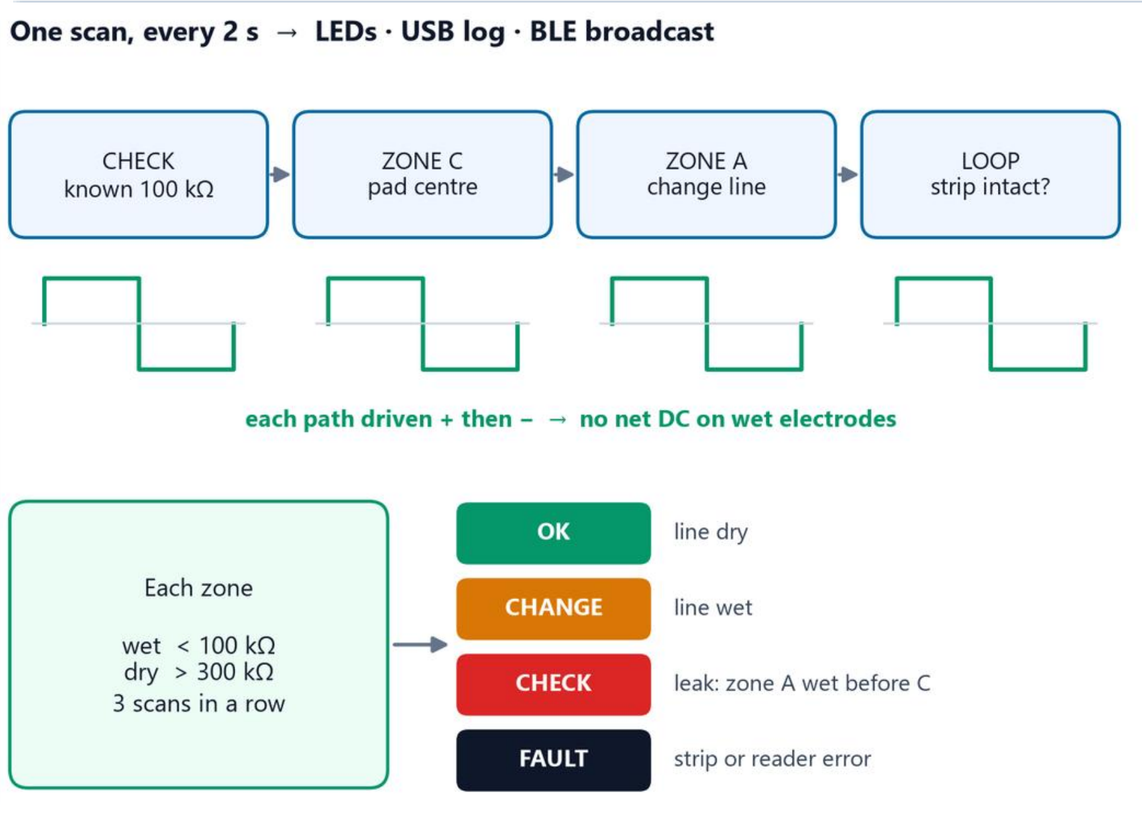
Design targets and hardware

Parameter	Design value	Basis
Sensing zones	2: centre C, change line A	decision D-012
Electrodes	1.0 mm lines and gaps	E2 pre-registered limits
Wet / dry	< 100 kΩ / > 300 kΩ, 3 scans	calc: wet 4–15 kΩ
Drive	bipolar, ≤ 33 µA, no net DC	calc · bench T8
ADC / MCU	ADS1115 16-bit (I²C) · ESP32-C3	datasheets [12]
Self-test	100 kΩ check ±5 % + strip loop	show FAULT if the check fails
Contacts	4 pogo pins + 2 dowels	CAD checked
Output	LEDs · USB log · BLE broadcast	BLE compiled only

Architecture Rationale: We are designing the reader for reuse; the strip is the disposable part.

System Architecture & Specs: Firmware & Data Flow

SOFTWARE CHECKS PASSED · HARDWARE TESTING PENDING



This diagram follows one reader scan from measurement to status.

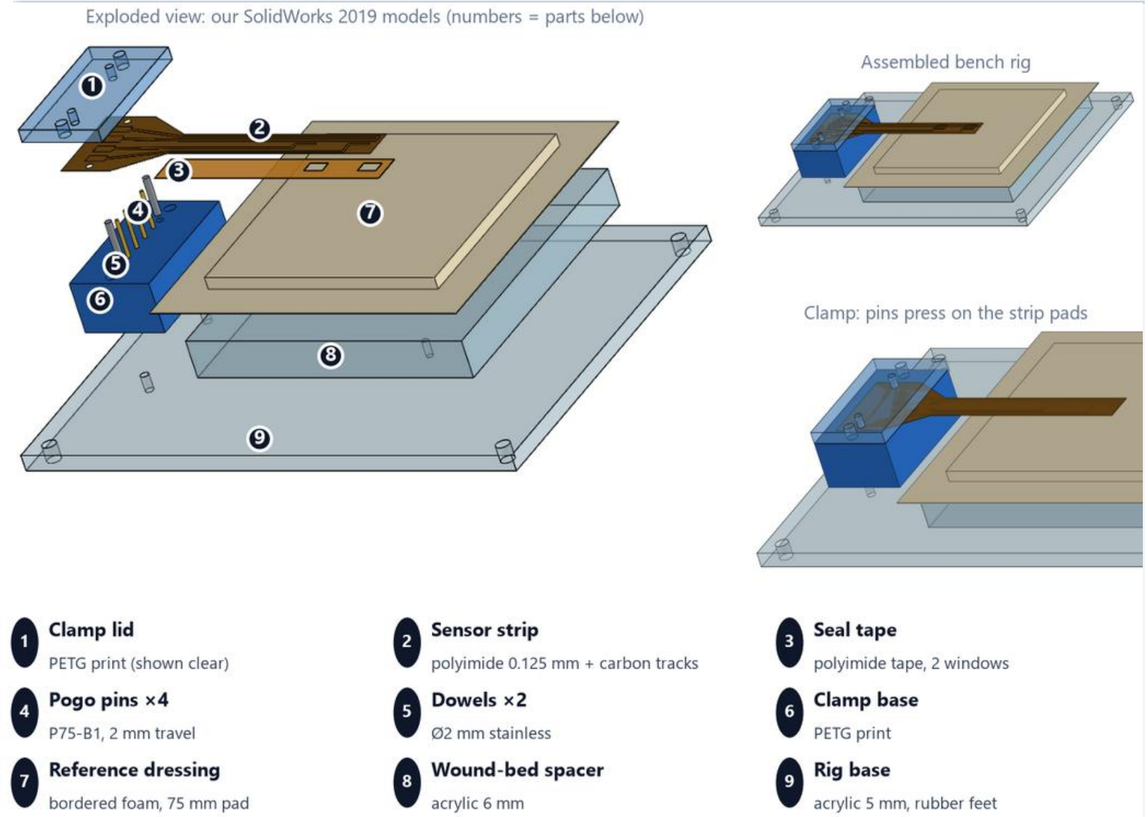
How the firmware turns readings into a status

- Measure: Each zone is driven in both directions to avoid net DC; hardware testing is pending.
- Read: The 16-bit ADC reading is scaled to the 3.3 V rail.
- Decide: Design thresholds are wet <100 kΩ and dry >300 kΩ for three consecutive scans.
- Self-check: A known 100 kΩ reference is checked each scan; a failed check shows FAULT.
- Output: Status appears on LEDs; data can be logged over USB. There is no custom app yet.
- Remote monitoring: the reader broadcasts its status over Bluetooth Low Energy (BLE) to any nearby smartphone scanner, with no hospital Wi-Fi or cloud needed. Firmware done; hardware test pending.

Software checks: We passed 23 compile-time checks and 16 PC-side tests. The firmware has not yet run on a wired board.

CAD Design & Mechanical Specs

DESIGNED · CAD VERIFIED · NOT YET BUILT



Our SolidWorks models and drawings; the clamp has not been built yet.

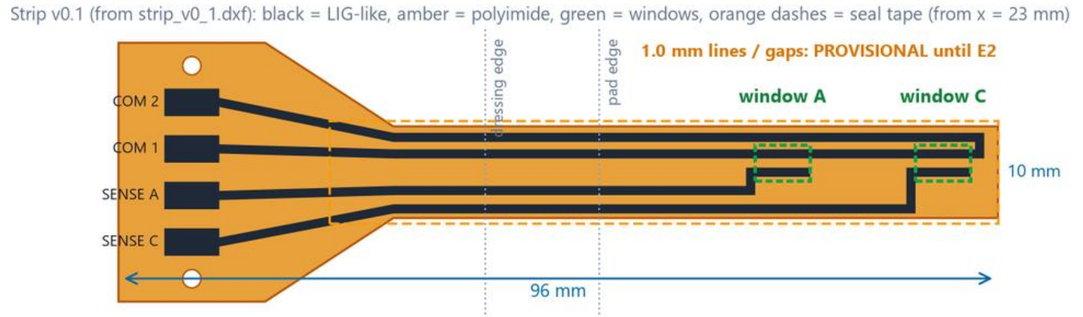
A clip-on clamp: 4 spring pins, 2 dowels

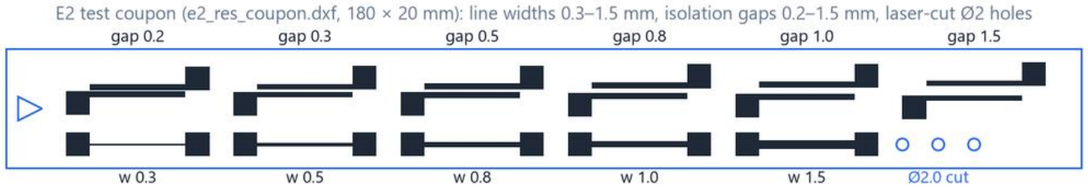
Check	Result	Basis
Materials	polyimide strip · PETG clamp · acrylic rig	printable, laser-cut
Pin force	0.72 N each · 2.9 N total	calc
Lid bending	11 µm vs 1.48 mm pin travel	calc
Alignment	0.49 mm used of ±1.35 mm	calc · 2.8× margin
Strip bending	0.62 % strain at 10 mm radius	calc · yield ≈ 3 %
CAD checks	99 passed · 0 failed · 5 blocked until E2 measurements	verify_cad.py
Stress check	lid stress ≈ 0.4 MPa · FoS > 40 on PETG	beam-model hand calc; FEA not run

Design Iteration: We found that the seal tape would sit under the clamp. We moved it to x = 23 mm; the CAD check now reports no clash (D-013).

CAD Design & Mechanical Specs: Sensing Strip & Laser Process

DESIGNED · LAB TEST PENDING





Laser pattern from our DXF files; it has not been fabricated yet.

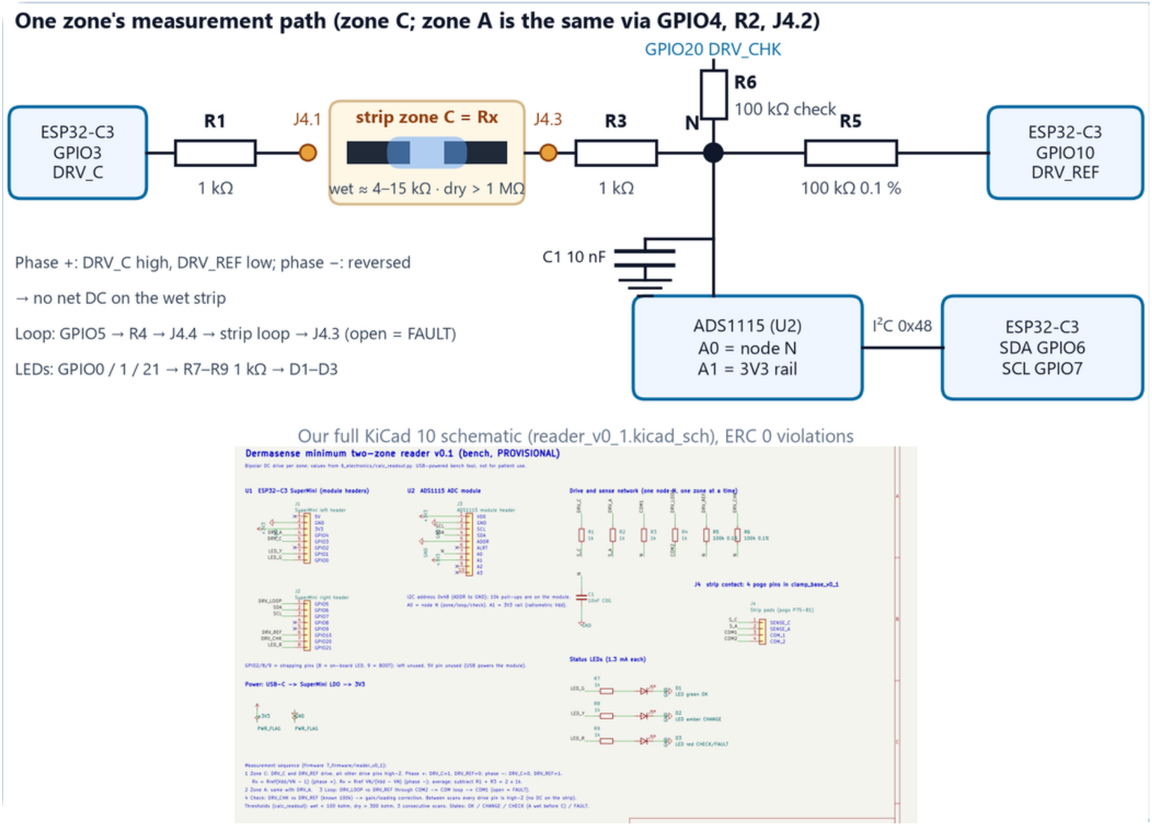
E2: laser and strip validation

- Material: We plan to create a laser-carbonised, LIG-like layer on polyimide [7].
- Reference: The machine's "12" setting is likely a PWM percentage, not watts; it still needs confirmation.
- E2 matrix: 7 coupons, 2 days, 72 pre-registered samples.
- Pass rules: Pre-set limits are CV ≤20% and electrical isolation across gaps >3 MΩ.
- Seal: laser-carbon made in air is superhydrophilic [9]; polyimide tape over the leads is designed to stop fluid wicking along the traces. E2 tests the seal.

All 35 design-file checks pass. No v0.1 strip has been cut yet; the LaserCAD import is confirmed at the machine before fabrication.

Electronics Architecture & Signal Path

SCHEMATIC CHECK PASSED · NOT YET WIRED



Our full KiCad 10 schematic (reader_v0_1.kicad_sch), ERC 0 violations

This figure shows zone C's measurement path from our KiCad 10 netlist.

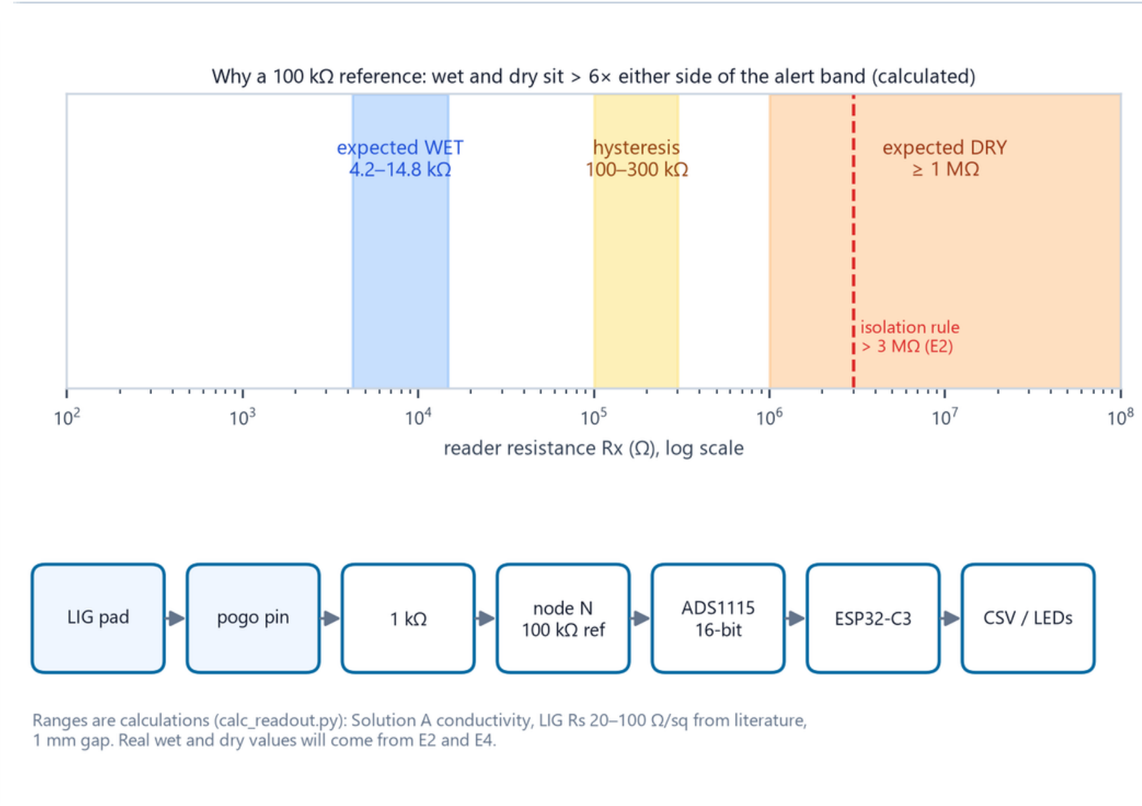
Pin and net map from our KiCad netlist

Net	Pin	Function
DRV_C · DRV_A	GPIO3 · GPIO4	drive zone C · A via 1 kΩ
DRV_LOOP	GPIO5	strip loop (broken strip)
DRV_REF	GPIO10	100 kΩ 0.1 % reference
DRV_CHK	GPIO20	100 kΩ self-check
SDA · SCL	GPIO6 · GPIO7	I²C to ADS1115 (0x48)
N · 3V3	ADS1115 A0 · A1	node · rail (ratiometric)
LED G · Y · R	GPIO0 · 1 · 21	OK · CHANGE · CHECK/FAULT
Strip pads	J4.1 – J4.4	S_C · S_A · COM1 · COM2
Power	USB-C → 3V3	LDO on the SuperMini

The electrical rules check found 0 violations. The netlist matches the firmware and strip pads, with 0 mismatches.

Electronics Schematics & BOM: Bill of Materials

3 ITEMS QUOTED · NOTHING ORDERED



Reusable reader ≈ ₹560–1,160 (BOM estimate) · Disposable strip: target ≤ ₹100

Bulk prototype purchases and test setup

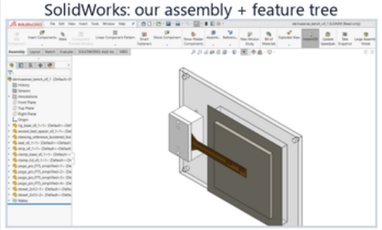
Item	Qty	Source	₹
ESP32-C3 SuperMini (PRO4607)	1	Probots · quoted	242
ADS1115 module (PRO789)	1	Probots · quoted	125
P75-B1 pogo pins (05880)	100	Hubtronics · quoted	354
Resistors, COG cap, LEDs, board	set	local / online	158–421
Clamp + bench rig	set	lab / local	100–1,240
Strip materials	set	lab / online	800–3,500
Test consumables	set	local	480–1,060
Prototype bulk outlay			**≈ 2,259–6,942**

Parts status: Three items quoted on 19 Oct; nothing ordered; other costs are estimates. Total ₹2,259–6,942 covers bulk purchases and bench/test setup, not one strip.

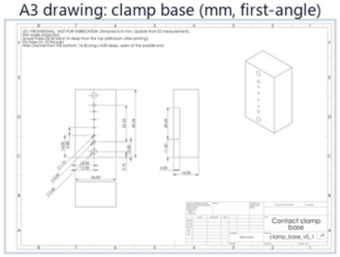
Maker Journey

PROJECT FILES · CHECKS CAN BE RERUN

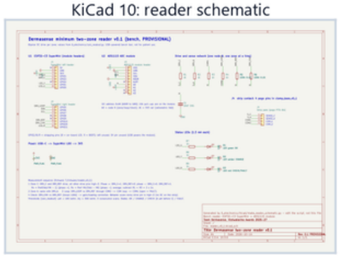
SolidWorks: our assembly + feature tree



A3 drawing: clamp base (mm, first-angle)



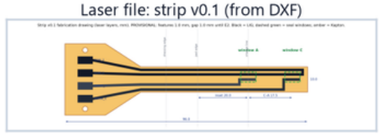
KiCad 10: reader schematic



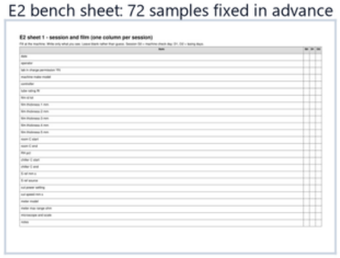
Firmware: state rule + compile-time test

```
// C = pad centre, A = maker's change line
constexpr State classify(Wet c, Wet a,
                         bool fault) {
  return fault ? State::Fault
    : (a == Wet::WET ? (c == Wet::WET
         ? State::Change : State::Check)
       : State::Ok);
}
static_assert(classify(Wet::WET, Wet::WET,
   false) == State::Change,
   "front reached the change line");
```

Laser file: strip v0.1 (from DXF)



E2 bench sheet: 72 samples fixed in advance

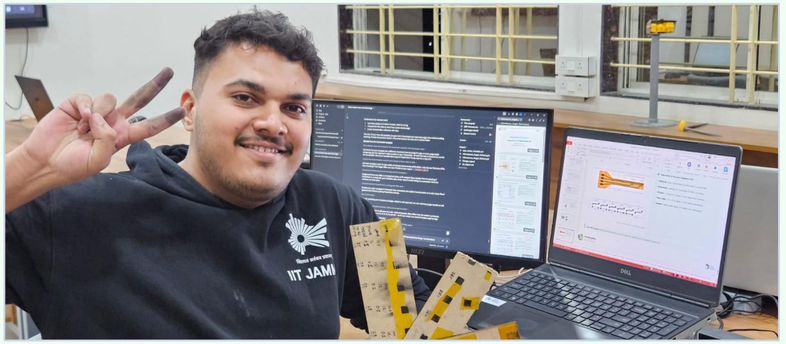


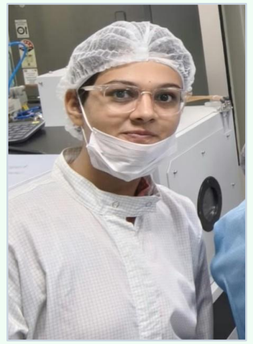
```
Checks re-run on 19 Oct 2026 (results as printed by each tool)

verify_cad.py (SolidWorks)          {'PASS': 99, 'FAIL': 0, 'BLOCKED': 5}
kicad-cli sch erc                   Found 0 violations
verify_laser_files.py               35 PASS, 0 FAIL
arduino-cli compile (esp32c3)       Sketch uses 650349 bytes (49%)
pytest (firmware tools)             16 passed
pytest (E2 analysis, synthetic data) 3 passed
```

Evidence from our project files and checks rerun on 19 October 2026.

Our team and project files





Soham Prabhakar Kadu · Lead, Mechanical Architecture & Sensor Design
Bhakti Ajay Kadam · Lead, Materials & Micro-fabrication (semiconductor background)

- Real files: The repository includes CAD models and drawings, the KiCad schematic, firmware and laser files, with dated work history since 14 October.
- Re-checkable: Project scripts can rerun the checks shown here.
- How we work: We used Claude to help write some scripts and reviewed their outputs. The physical build and tests are still planned, starting with E2.
- Photos: real, unedited photos of our team: Soham with early laser-written test samples; Bhakti in the cleanroom.

Project files: [github.com/soham0777/dermasense-vishwakarma-2026 · native CAD, KiCad, firmware and laser files with dated work history.]

Maker Journey: Engineering Iterations

DECISIONS RECORDED IN OUR PROJECT

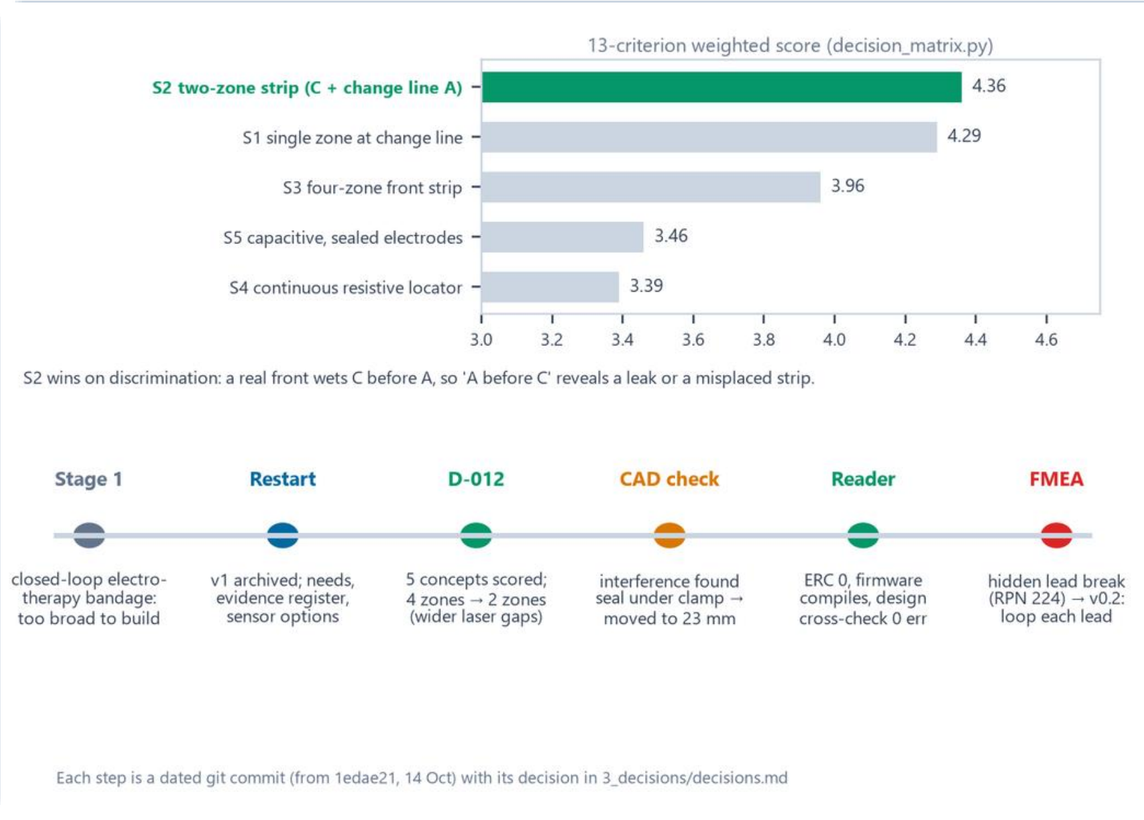


Figure: concept scores and the iteration path (each a dated commit).

How our design changed from Stage 1 to Stage 2

- Pivot: We narrowed the idea from a therapy bandage to one question: has the dressing reached its change line?
- Choice: two zones (score 4.36) beat one zone (4.29), because only two zones give the order check (centre before change line) that flags leaks.
- Kept: We kept Bluetooth as a local status broadcast. It compiles; hardware testing is pending.
- Corrected: We corrected the Stage 1 description of Kapton: DuPont cautions against calling it medical-grade [11].
- Redesign: To address the FMEA risk, version 0.2 loops each sensing lead.

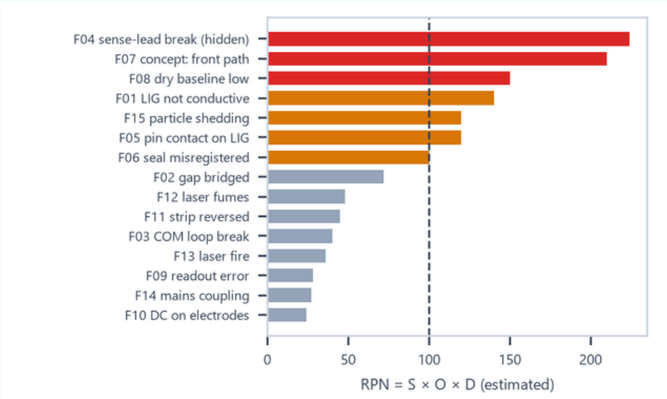
Existing options: diaper strips[4] and visual change indicators[14] already exist. We are testing a two-zone strip that reads the change line electrically and checks which zone wets first.

Safety & Hazard Mitigation

RISKS ESTIMATED · TESTS PLANNED

We ranked 15 possible failure modes before testing. These are the five highest estimated risks.

Failure mode	Effect	RPN	Fix / test
F04 cracked sense lead	zone reads 'dry' forever; no alert	224	v0.2 loops each lead · T-MECH-01
F07 front path differs	CHANGE late or missing	210	E4 on real foam dressings
F08 humid dry baseline	humidity reads as 'wet'	150	measure baselines (E4)
F01 weak carbon layer	both zones always read 'dry'	140	E2 + per-strip QC record
F05 pin contact on carbon	readings drift high	120	rounded pins · T-CON-01



Fail-safes

- **Bench only:** Use the USB reader on the bench only; do not put it on a person.
- **No net DC:** The drive is bipolar and idle pins are off. Planned bench test T8 checks for less than 5 mV residual.
- **Self-check:** Run a self-check each scan; show FAULT if it fails.
- **No wound contact:** The strip is designed to sit on the pad's outer face, with sealed leads.
- **Laser work:** Complete and sign the laser-safety checklist before each session.

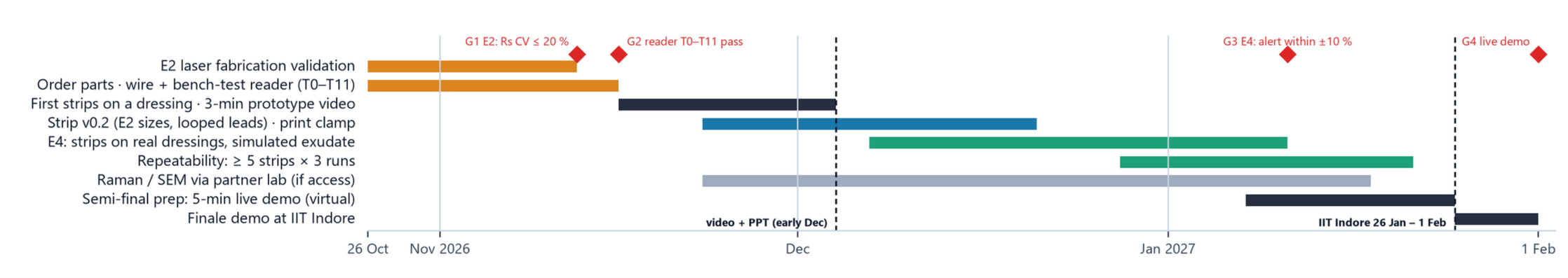
Standards (considered, not claimed)

- **ISO 14971** risk management
- **IEC 60601-1** future on-body reader
- **ISO 10993-1** before any skin contact[11]
- **EN 13726** simulated exudate for tests[10]
- **IEC 60825-1** laser work during fabrication

Roadmap

PLANNED WORK · GATED MILESTONES

Four-month plan aligned with the programme dates. Each step depends on passing the previous gate.



Build & Bench (Months 1–2)

- Nov: Run E2 laser tests (G1), then build and bench-test the reader (G2).
- By 4 Dec: If gates pass, complete the first dressing demo, prototype video and revised Stage 3 deck.

Validate & Demo (Months 3–4)

- Dec–Jan: Build strip version 0.2 and a printed clamp, then run E4 wet tests (G3, mid-Jan) if earlier gates pass.
- Jan: Prepare for virtual jury interviews.
- If selected, Grand Finale at IIT Indore (26 Jan–1 Feb 2027): live jury demo of the strip and reader on a dressing, showing OK → CHANGE.

Technical Challenges & Risks

ISSUES FOUND DURING OUR CHECKS

Geometry: seal tape under the clamp

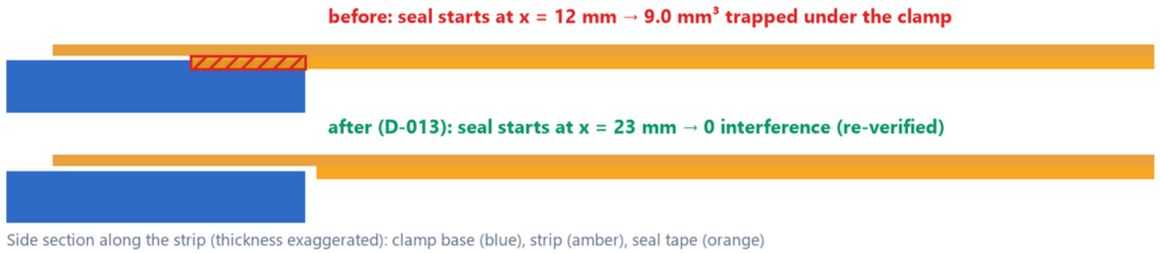
Found: We found the seal tape overlapped the clamp by 9.0 mm³.
Impact: This could lift the strip off the contact pins.
How: We found it with an automated CAD clash check.

Circuit: a broken lead can look dry

Found: A cracked sensing lead could make the reader report “dry”.
Impact: The CHANGE status might never appear, so there would be no alert.
How: The FMEA ranked this risk at 224.

Process: low-PWM tube current stability

Found: The machine’s “12” setting is likely a PWM percentage, not a current measurement. Laser output can be unstable at low power [8].
Impact: trace widths are held at a conservative 1.0 mm until E2 calibrates the laser setting (and the tube current, if a meter is available).
How: We found the uncertainty while checking the LaserCAD configuration.



Side section along the strip (thickness exaggerated): clamp base (blue), strip (amber), seal tape (orange)



We also corrected two drawing issues: STEP files now export in millimetres, and a 0.125 mm dimension is shown correctly.

Potential Solutions & Risk Mitigation

CHECKED · DESIGNED · TO TEST

Checked in CAD or software

Seal: starts at x = 23 mm; 0 clash.
Units: all STEP files in mm.
Checks: 99 checks passed, 0 failed; 5 await E2 measurements.

Designed, not yet tested

v0.2 strip: every sense lead looped.
Contact: silver pads, rounded pins.
Self-test: check resistor → FAULT.

Fallbacks

Wider lines: if E2 says so.
Process: slower passes, log current.
Concept: one-zone strip (score 4.29).

What we still need to test

- **Experiments:** Run E2 laser tests, bench-test the reader, then run E4 wet tests.
- **Material:** Use Raman and SEM before describing the material as graphene.
- **Reliability:** Test bending, contact wear and humidity-related drift.
- **Product:** Explore a battery-powered reader and a secure local phone connection. Neither is built yet; one strip is intended per dressing.
- **Before any study with people: address relevant biocompatibility and electrical-safety requirements and obtain ethics approval.**
- **Not claimed:** We have not established sensor performance, graphene identity or clinical benefit.

Our rule: we will not claim sensor performance, graphene identity or clinical benefit until tests support those claims.

References & Sources

SOURCES AND WHAT WE READ

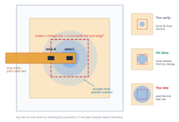
Sources are numbered below. Reading depth is noted; a citation does not mean a claim was independently verified.

1. Milne S.D. et al., A wearable wound moisture sensor as an indicator for wound dressing change, Int. Wound J. 13(6):1309–1314 (2016). doi:10.1111/iwj.12521 · full text read
2. Dressing instructions for use: some foam dressings say 'change when exudate approaches ~2 cm from the pad edge'; Mölnlycke Mepilex Border Flex brochure uses strike-through reaching 3 edges · manufacturer documents, read as excerpts
3. McColl D., Cartlidge B., Connolly P., Real-time monitoring of moisture levels in wound dressings in vitro, Int. J. Surg. 5(5):316–322 (2007). doi:10.1016/j.ijsu.2007.02.008 · abstract
4. US 5,469,145 and US 5,838,240 (wet diaper detector: detachable strip + clip-on detector); US 9,291,589; US 10,928,344 · patent summaries, claims not read
5. Jiang Y. et al., Wireless, closed-loop, smart bandage with integrated sensors and stimulators, Nat. Biotechnol. (2022). doi:10.1038/s41587-022-01528-3 · listing + coverage
6. Dealey C., Cameron J., Arrowsmith M., A study comparing two objective methods of quantifying the production of wound exudate, J. Wound Care 15(4):149–153 (2006) · abstract (methods disagree)
7. Lin J. et al., Laser-induced porous graphene films from commercial polymers, Nat. Commun. 5:5714 (2014). doi:10.1038/ncomms6714 · full text read
8. CO_2 laser tube current vs PWM setting, engineering measurements, softsolder.com (2022) · not peer-reviewed
9. Li Y. et al., Laser-induced graphene in controlled atmospheres: from superhydrophilic to superhydrophobic surfaces, Adv. Mater. 29:1700496 (2017). doi:10.1002/adma.201700496 · abstract
10. EN 13726-1:2002 / EN 13726:2023, Test methods for wound dressings (absorbency; 'Solution A') · catalogue entry; online link not reachable from our network
11. DuPont position statement on the use of DuPont materials in medical applications (Kapton caution) · web page
12. Texas Instruments ADS1115 datasheet (SBAS444); Espressif ESP32-C3 datasheet · vendor documents
13. Wounds UK news (22 Aug 2018) on a Mölnlycke-commissioned survey of >200 UK patients with chronic wounds: average dressing time 18 min · patient-reported, industry-sponsored; web page read
14. Smith+Nephew, ALLEVYN LIFE product page: visual change indicator ('At 75%, it's time to change your dressing') · manufacturer web page, read 19 Oct 2026
15. Truemeds.in retail listings: Mepilex Lite 10 × 10 cm ₹242; Mepilex Border 10 × 10 cm ₹514 (pack size unclear) · retail pages, read 19 Oct 2026
16. Gupta N., Gupta S.K., Shukla V.K., Singh S.P., An Indian community-based epidemiological study of wounds, J. Wound Care 13(8):323–325 (2004). doi:10.12968/jowc.2004.13.8.26657 · abstract
17. Sharma A. et al., Epidemiological profile of chronic wounds in an Indian population, Cureus 16(9):e68684 (2024). doi:10.7759/cureus.68684 · full text
18. Anjana R.M. et al., ICMR-INDIAB national cross-sectional study (ICMR-INDIAB-17), Lancet Diabetes Endocrinol. 11(7):474–489 (2023) · abstract and summaries
19. Armstrong D.G., Boulton A.J.M., Bus S.A., Diabetic foot ulcers and their recurrence, N. Engl. J. Med. 376:2367–2375 (2017). doi:10.1056/NEJMra1615439 · summaries; range is an estimate and contested
20. Price P.E. et al., Dressing-related pain in patients with chronic wounds: an international patient perspective, Int. Wound J. 5(2):159–171 (2008). doi:10.1111/j.1742-481X.2008.00471.x · abstract
21. Lavery L.A. et al., WHS guidelines update: diabetic foot ulcer treatment guidelines, Wound Repair Regen. (2024). doi:10.1111/wrr.13133 · full text

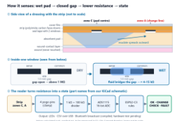
Where the figures came from

FIGURES MADE FROM OUR PROJECT FILES

The illustrations come from our CAD, schematic exports and plots made from project files. We did not use stock images or AI image generators. Team photos on slide 12 are the team's own.



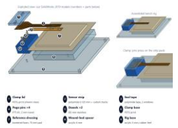
Problem illustration, drawn to scale
Source: make_figures.py · strip_v0_1_geometry.json
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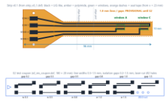
How it senses (concept)
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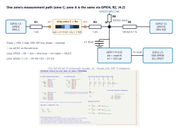
Firmware scan and state logic
Source: make_figures.py · reader_v0_1 source
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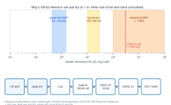
Annotated SolidWorks views
Source: render_cad.py + make_figures.py · 5_cad/solidworks
1_presentation/figures/fig_cad.png



Strip and E2 coupon layers
Source: make_figures.py · our DXF files
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Measurement path + KiCad sheet
Source: make_figures.py · reader_v0_1 netlist + KiCad PDF
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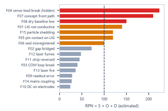
Measurement window
Source: make_figures.py · calc_readout.py
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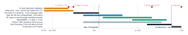
Evidence gallery + tool output
Source: our files, screen capture, 19 Oct tool runs
1_presentation/figures/fig_maker.png



Decision matrix + iterations
Source: make_figures.py · decision_matrix.md
1_presentation/figures/fig_decision.png



FMEA risk ranking
Source: make_figures.py · fmea_v0_1.csv
1_presentation/figures/fig_fmea.png



4-month roadmap
Source: make_figures.py · E2/bench/E4 plans
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Seal clash + lead-break diagrams
Source: make_figures.py · geometry + FMEA
1_presentation/figures/fig_challenges.png